

6CS012 – Artificial Intelligence and Machine Learning.

Lecture – 02



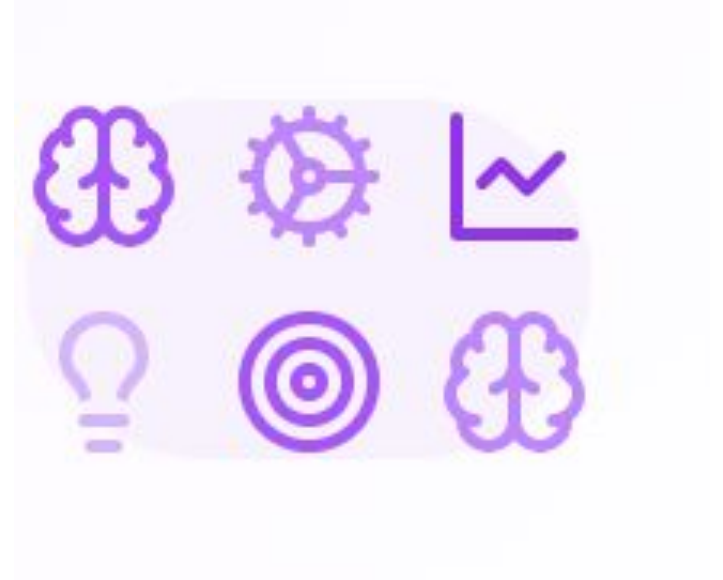
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Understanding the Components of Learning.

A Classification Perspective.

Learning Objectives:

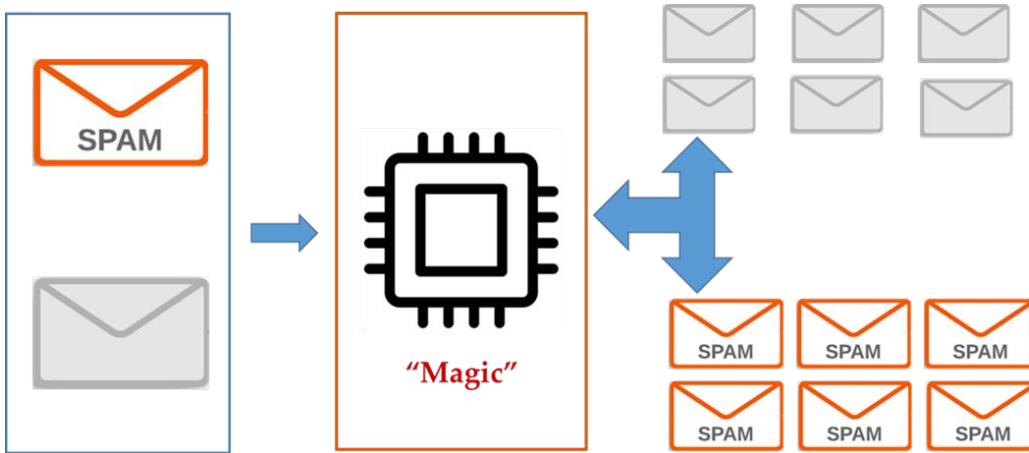
- Review key Machine Learning concepts from 5CS037.
- Understand Logistic Regression for classification tasks.
- Identify limitations of Logistic Regression.
- Justify the need for Deep Learning.



What is Learning?

The Learning Problems.

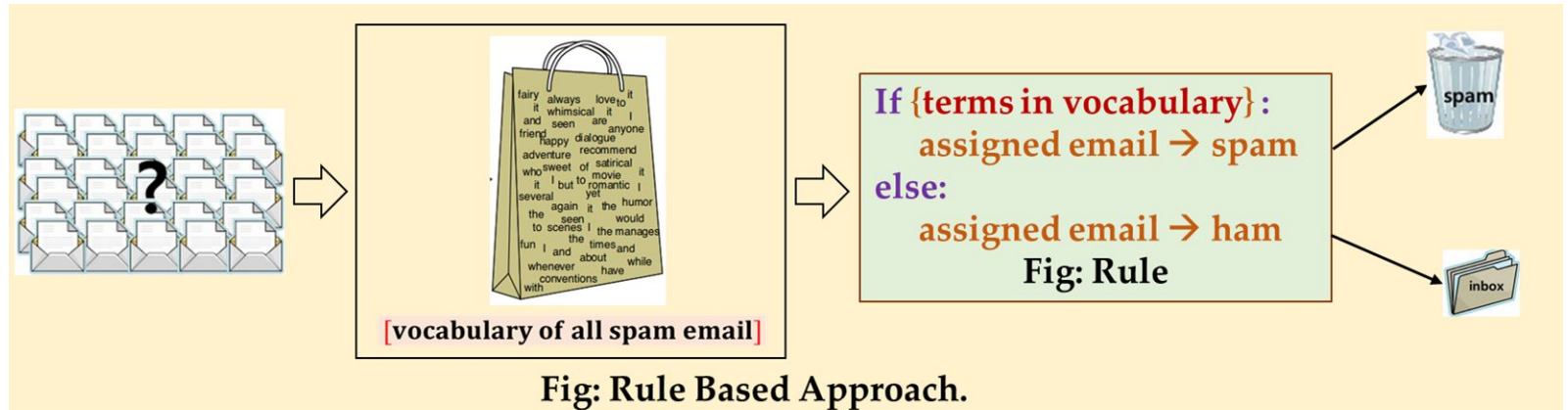
- Task-Example: Identify the spam emails!!!
 - (Program a machine that learns how to filter spam emails.)



Let's Demystify the Magic!!



The Expert System – Rule Based.



Limitations of Expert System

- **Interpretability:** Rules are manually crafted and limited to specific domains.
- **Static:** Cannot adapt to changing logic; rewriting required.
- **Shallow:** Lacks deep understanding, relies on memorizing features of data.

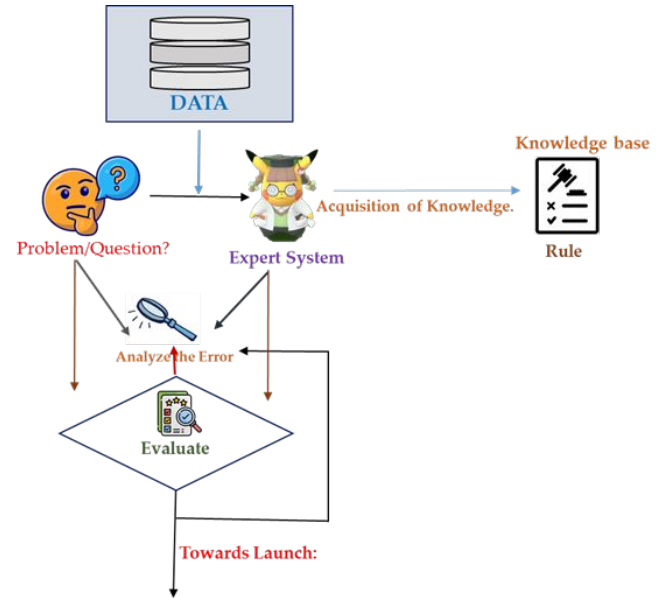


Fig: A basic representation of Expert System.

Learning from Data – Machine Learning

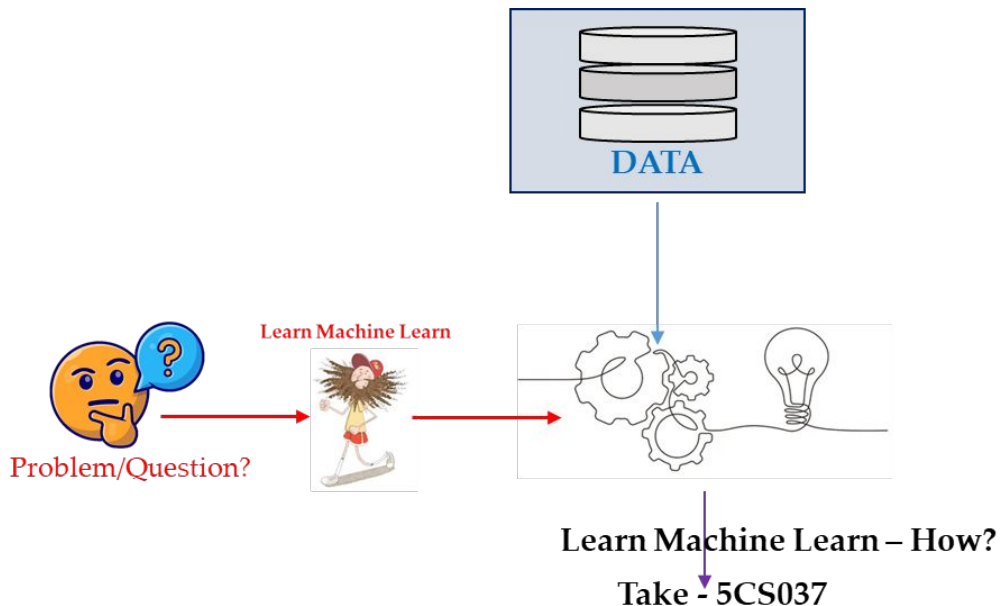
Instead of manually defining decision rules, machines **learn** from data:

Training data: Pairs of inputs (x) and outputs (y).

Examples

Videos labeled as "cat" or "no cat."

Emails labeled as "spam" or "not spam."



What is Machine Learning?

- What Legend has to say about it?

- “Learning is any process by which a system improves performance from experience”.

- Herbert Simon



- “Field of study that gives computers the ability to learn without being explicitly programmed.”

- Arthur Samuel, 1959 (an AI pioneer at IBM and founding fellow of Association for the advancement of Artificial Intelligence).



- Definition by Tom Mitchel(1998):

- Machine Learning is the study of algorithms that:

- Improve their performance P
 - At some task T
 - With experience E

- A well defined learning task is given by $\langle P, T, E \rangle$.



Tasks that Require Machine Learning.

- What makes a 2?
- What distinguishes a 2 from 7?



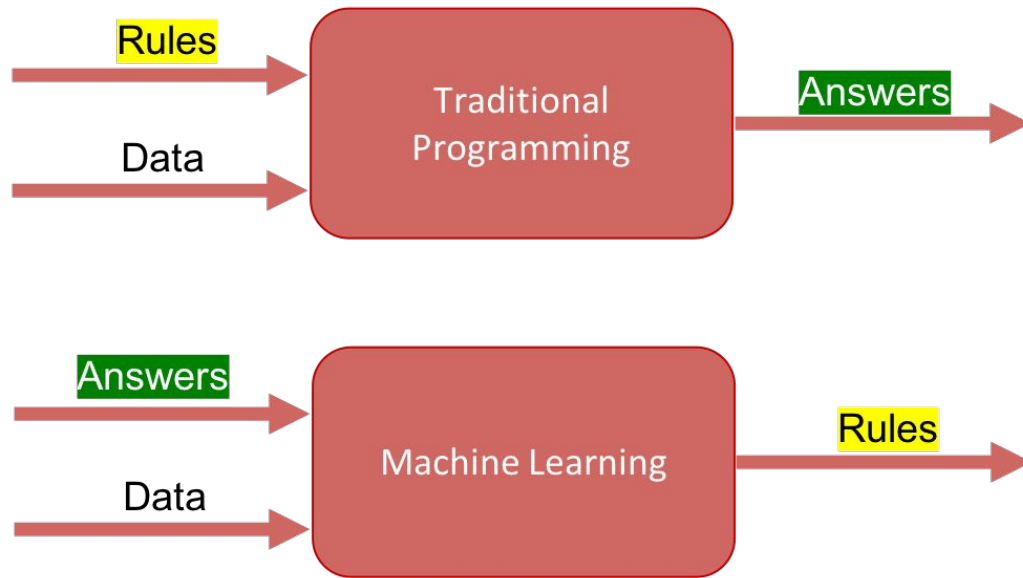
In Mitchel Tom Definition:

T: Recognize hand written digits.

P: Percentage of digit recognized correctly.

E: Database of human-labeled images of handwritten digits.

Machine Learning Vs. Traditional Programming.



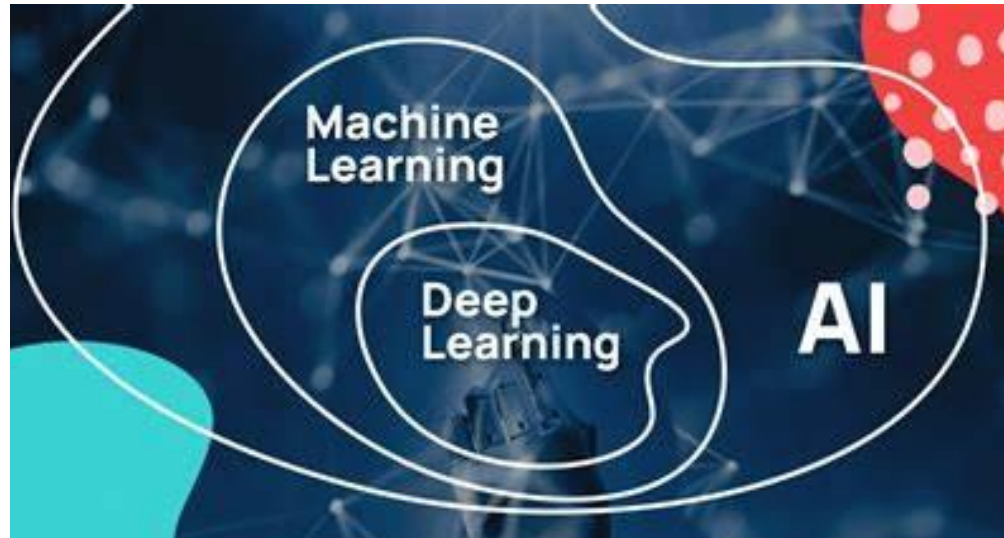
What isn't Machine Learning?

Powerful but not magical!

Not *Exactly* Artificial Intelligence

Limitations:

- Not a path to robot singularity
- Struggles with abstract reasoning
- Not understanding, just processing

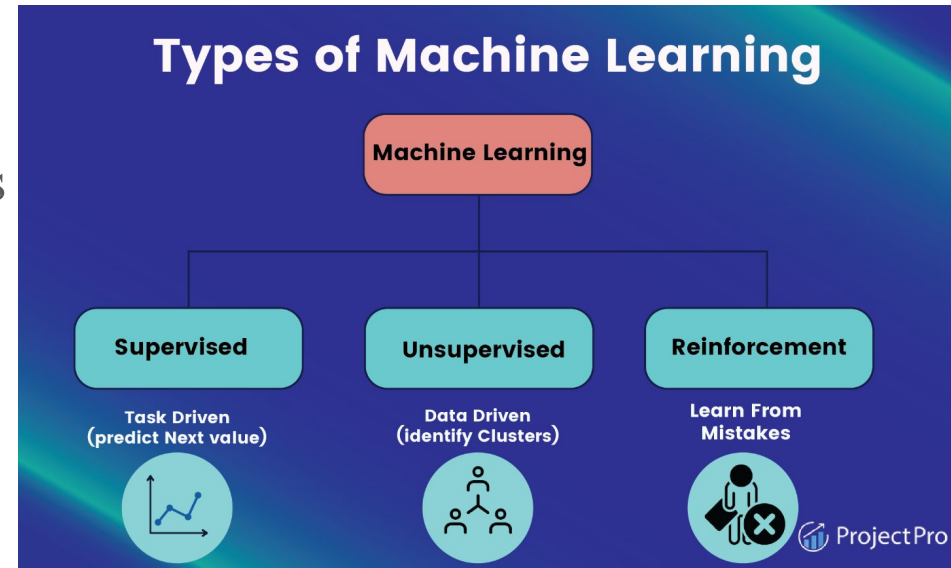


Core Components in Machine Learning



Types of Machine Learning

- Many different problem classes in machine learning.
- Vary according to what kind of data is provided and what kind of conclusions are to be drawn from it
- Some popular kinds
 - Supervised Learning
 - Unsupervised Learning



Supervised Machine Learning

- Uses Labeled Dataset
- Types
 - Classification
 - Regression

	pregnant	glucose	bp	skin	insulin	bmi	pedigree	age	label
1	6	148	72	35	0	33.6	0.627	50	1
2	1	85	66	29	0	26.6	0.351	31	0
3	8	183	64	0	0	23.3	0.672	32	1
4	1	89	66	23	94	28.1	0.167	21	0
5	0	137	40	35	168	43.1	2.288	33	1
...	...	...	...	...	...	...	...	...	...
764	10	101	76	48	180	32.9	0.171	63	0
765	2	122	70	27	0	36.8	0.34	27	0
766	5	121	72	23	112	26.2	0.245	30	0
767	1	126	60	0	0	30.1	0.349	47	1
768	1	93	70	31	0	30.4	0.315	23	0

768 rows × 9 columns

Regression

- Predicts continuous numerical values
- Finds relationship between input features and target variable

Input: Numerical features

Output: Continuous numeric prediction

Goal: Minimizes prediction error

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Street	OverallCond	YearBuilt	YearRemd	MasVnrArea	TotalBsmf	Heating	CentralAir	BsmfFullB	FullBath	HalfBath	Bedroom	SaleCondition	Price
1 Pave	6	1961	1961	0	882	GasA	Y	0	1	0	2	Normal	11622
1 Pave	6	1958	1958	108	1329	GasA	Y	0	1	1	3	Normal	14267
1 Pave	5	1997	1998	0	928	GasA	Y	0	2	1	3	Normal	13830
1 Pave	6	1998	1998	20	926	GasA	Y	0	2	1	3	Normal	9978
1 Pave	5	1992	1992	0	1280	GasA	Y	0	2	0	2	Normal	5005
1 Pave	5	1993	1994	0	763	GasA	Y	0	2	1	3	Normal	10000
1 Pave	7	1992	2007	0	1168	GasA	Y	1	2	0	3	Normal	7980
1 Pave	5	1998	1998	0	789	GasA	Y	0	2	1	3	Normal	8402
0 Pave	5	1990	1990	0	1300	GasA	Y	1	1	1	2	Normal	10176
1 Pave	5	1970	1970	0	882	GasA	Y	1	1	0	2	Normal	8400
2 Pave	5	1999	1999	0	1405	GasA	Y	1	2	0	2	Normal	5858
3 Pave	5	1971	1971	504	483	GasA	Y	0	1	1	2	Normal	1680
4 Pave	5	1971	1971	492	525	GasA	Y	0	1	1	3	Normal	1680
5 Pave	6	1975	1975	0	855	GasA	Y	0	2	1	3	Normal	2280
6 Pave	6	1975	1975	0	836	GasA	Y	0	1	0	2	Normal	2280
7 Pave	5	2009	2010	162	1590	GasA	Y	0	2	1	3	Partial	12858
8 Pave	5	2009	2010	256	1544	GasA	Y	0	2	0	3	Partial	12883
9 Pave	5	2005	2005	615	1698	GasA	Y	0	2	0	3	Normal	11520
0 Pave	5	2005	2006	240	1822	GasA	Y	0	2	0	3	Normal	14122
1 Pave	5	2003	2004	1095	2846	GasA	Y	1	2	1	3	Normal	14300
2 Pave	5	2002	2002	232	1671	GasA	Y	1	2	1	3	Normal	13650
3 Pave	5	2006	2006	178	1370	GasA	Y	0	2	0	2	Normal	7132
4 Pave	5	2005	2005	0	1324	GasA	Y	0	2	0	3	Normal	18494
5 Pave	5	2006	2006	14	1145	GasA	Y	0	2	0	2	Normal	3203
6 Pave	5	2004	2004	0	384	GasA	Y	1	2	1	3	Normal	13300

Classification

- Predicts discrete class labels
- Maps input data to predefined categories

Input: Features (attributes)

Output: Discrete class/category

Learns from labeled training data

	pregnant	glucose	bp	skin	insulin	bmi	pedigree	age	label
1	6	148	72	35	0	33.6	0.627	50	1
2	1	85	66	29	0	26.6	0.351	31	0
3	8	183	64	0	0	23.3	0.672	32	1
4	1	89	66	23	94	28.1	0.167	21	0
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What do we want?

Learning

Finding patterns in training data to create a model that makes predictions.

Inference

Using the model to predict and interpret outcomes for new data.



Models

A model (or hypothesis) is a function that maps feature vectors to predicted target values

$$y = f(X; \theta)$$

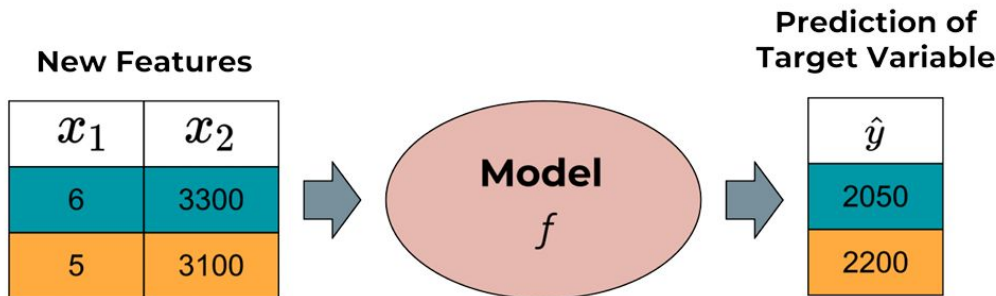
Where:

y : Target Variable

X : vector of input features

Θ : Parameters that needs to be learned from the data

f is meant to capture intrinsic patterns of the data, the underlying assumption being that these hold true for all data drawn from P_{xy}

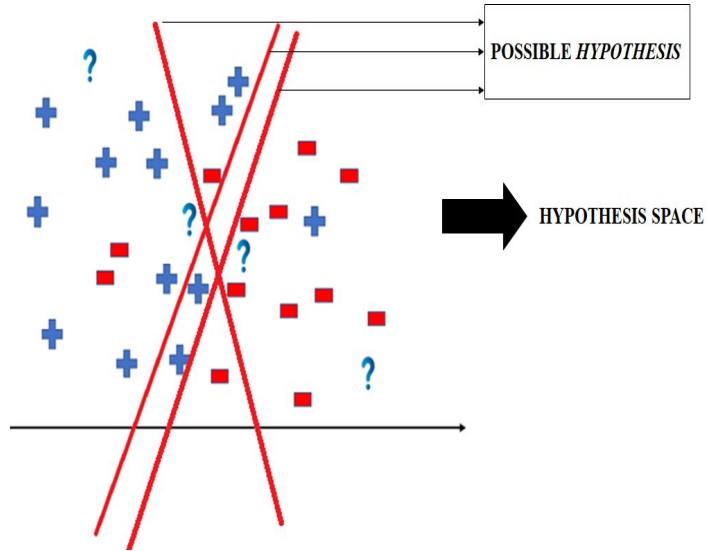


Models in Hypothesis Space

To find a "good" model, we must first decide on a specific type of model. This limits the options and makes the task manageable.

This predefined type is called a **structural prior**. The set of functions for this model type is the **hypothesis space (H)**:

$H = \{f: f \text{ belongs to a specific functional family } y\}.$



Parameterization

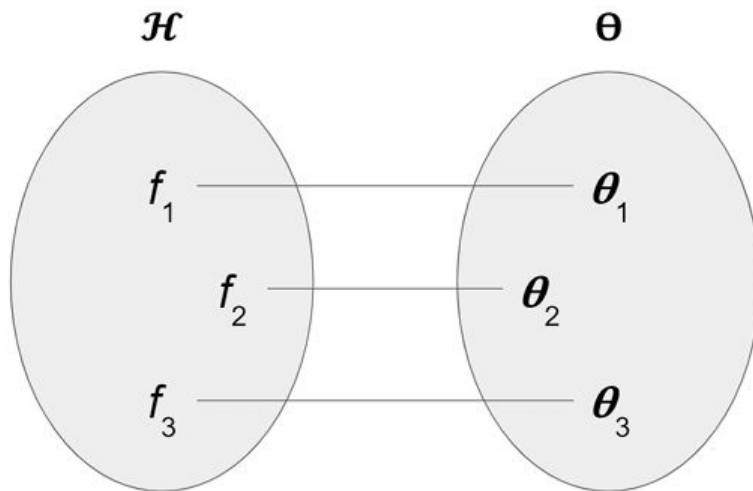
- We collect all parameters in a parameter vector

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_d)$$

from parameter space Θ .

- So,

Finding the optimal model is perfectly equivalent to finding the optimal set of parameter values.



How?

- How do we fit or train the model so we get best set of parameters?
- How do we determine which set of parameters best fits our function or model?

Cost Function

A cost function measures how well a model's predictions match the actual target values. It quantifies the error.

Goal: Find parameters that minimize the loss (Empirical Risk Minimization)

$$\ell(f(x, y)) = \frac{1}{N} \sum^N \ell_i(f(x_i), y_i)$$

A big part of machine learning focuses on the question, **how to do this minimization efficiently?**

Optimization Process

Optimization Problem:

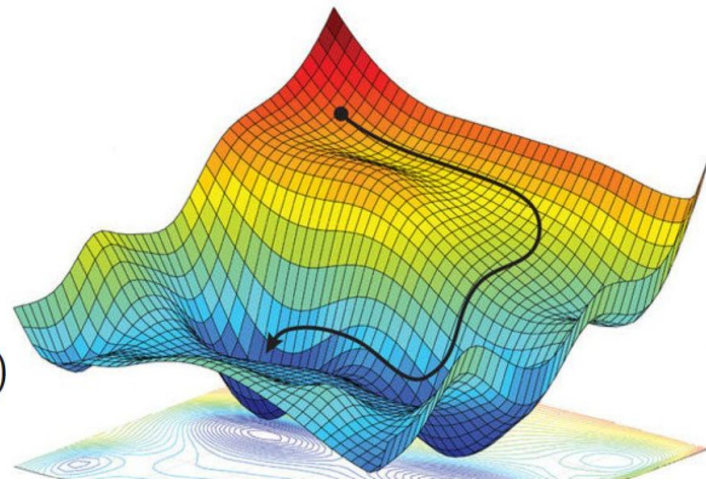
Find $\hat{f} = \arg \min(f)$, minimizing a **loss function** ℓ .

(Also called an objective function.)

ML as Optimization:

Find parameters θ^* minimizing $\ell(f(x_i, y_i))$:

$$\theta^* = \arg \min_{\theta \in \Theta} \ell(x, y, \theta, f)$$

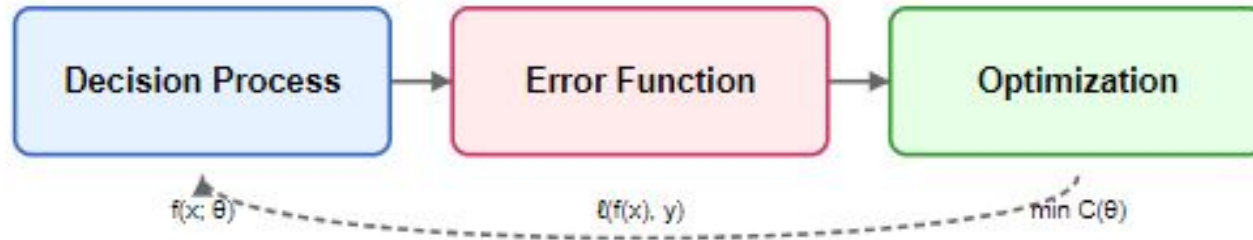


Methods to Solve for θ^* :

- Analytic solutions
- Random search.
- Systematic search.
- Gradient-based numeric methods.

Summary: Machine Learning (ERM framework)

- Decision Process/Function
- Error Function
- Optimization Process



$$C(\theta) = (1/N) \sum \ell(f(x_i), y_i)$$

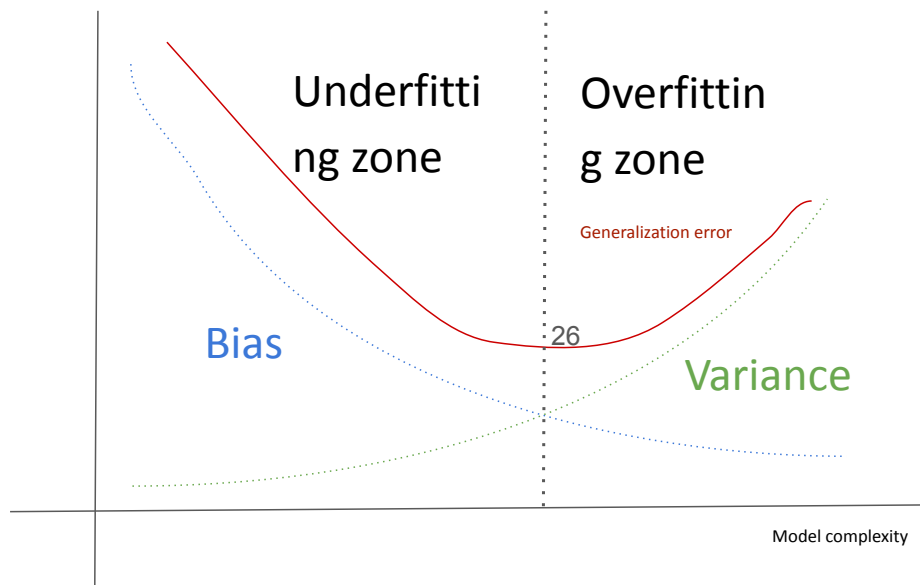
How good is your model?

Idea of Generalization!

Bias Variance Tradeoff

- Models with high bias will have low variance.
- Models with high variance will have a low bias.

The idea is to find a “**sweet spot**”.

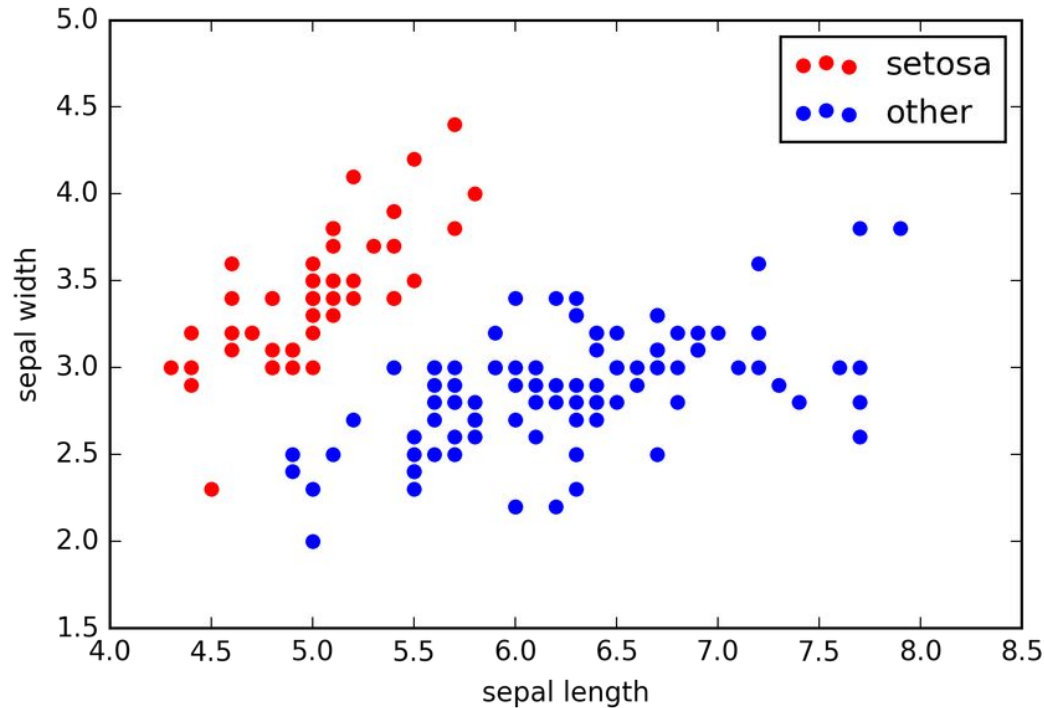


Understanding Classification Task

(With Logistic Regression Algorithm)
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Linear Models

- Separate classes using a linear decision boundary
- Simple, interpretable, and computationally efficient
- Common examples: Logistic Regression, SVM etc



Classification

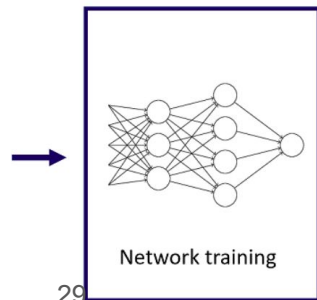
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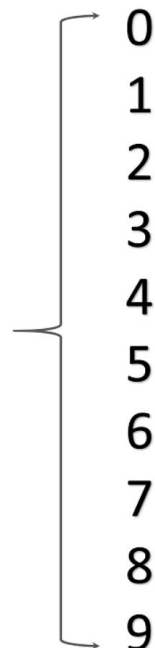
Binary



Data & Labels



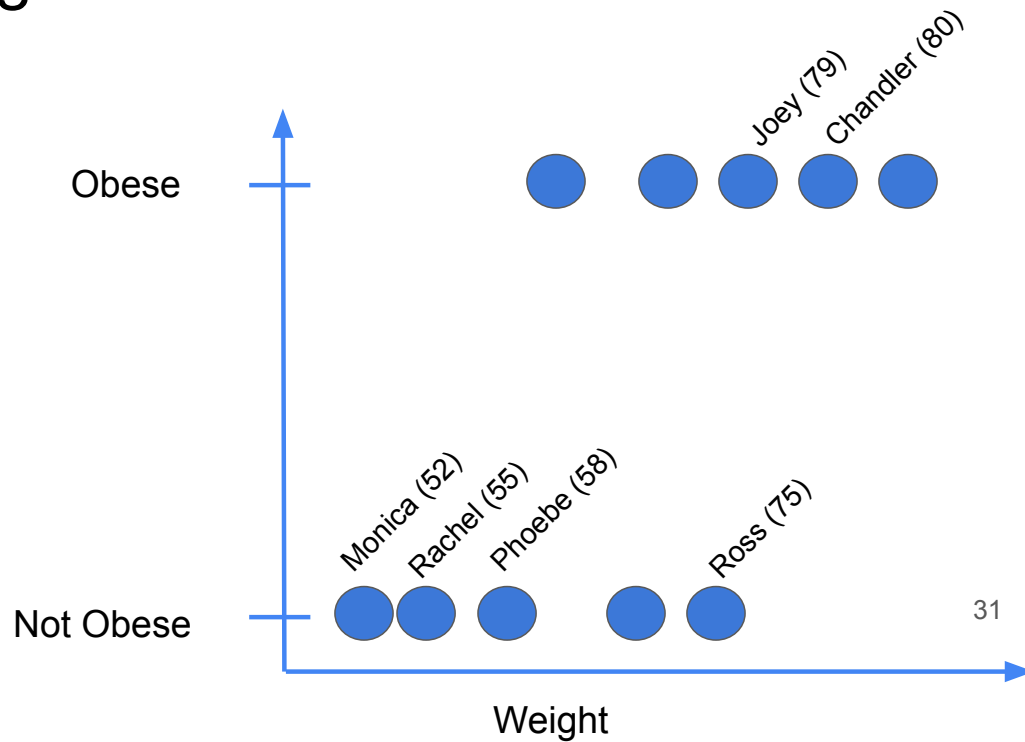
Multi-class



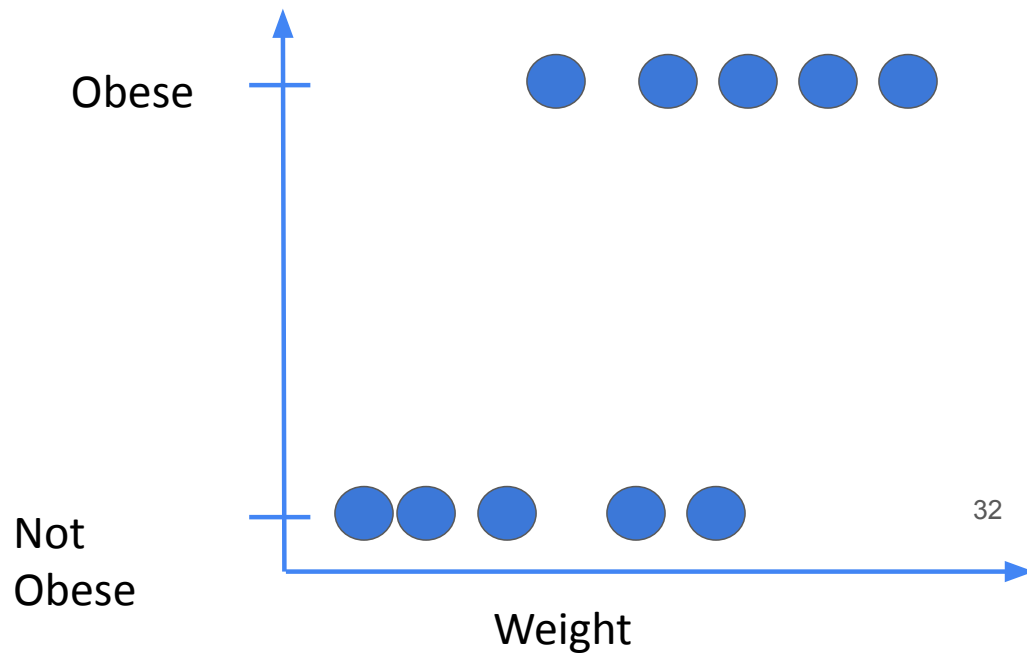
Take a look at this example

Name	Weight	Is Obese?
Chandler	80	1
Monica	52	0
Phoebe	58	0
Rachel	55	0
Ross	75	0
Joey	79	1

Logistic Regression Motivation



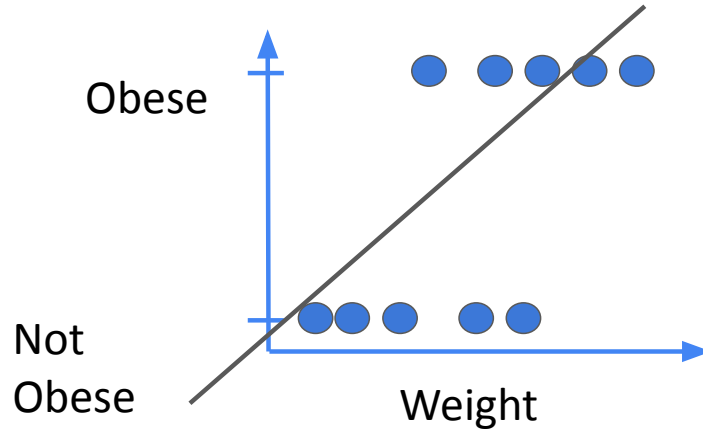
Logistic Regression Motivation



In Linear Model

We considered using a linear regression model to represent these probabilities

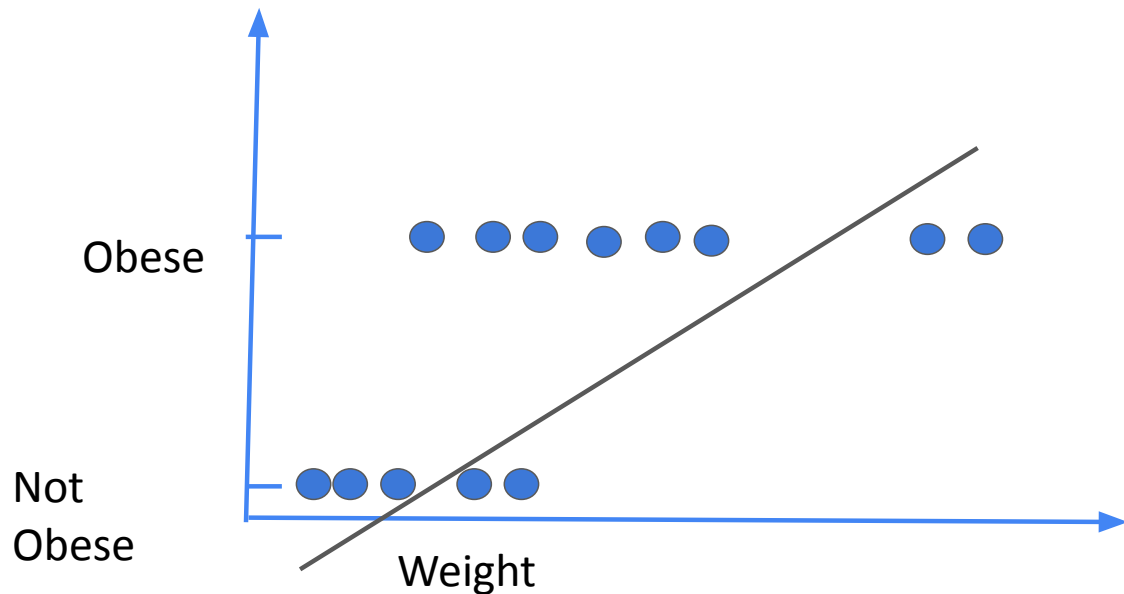
$$p(X) = \beta_0 + \beta_1 X$$



Problem with linear model (In the classification task)

We considered using a linear regression model to represent these probabilities

$$p(X) = w_0 + w_1 X$$



The Logistic Model

The Logistic Model

Rather than modeling the response Y directly, the logistic regression models the probability that Y belongs to a particular category.

$\Pr(\text{obese} = \text{Yes} \mid \text{weight})$

One might predict $\text{obese} = \text{Yes}$ for any individual for whose $p(\text{weight}) > 0.5$.

For a more conservative approach, you may choose to use a lower threshold, such as $p(\text{weight}) > 0.1$.

The Logistic Model

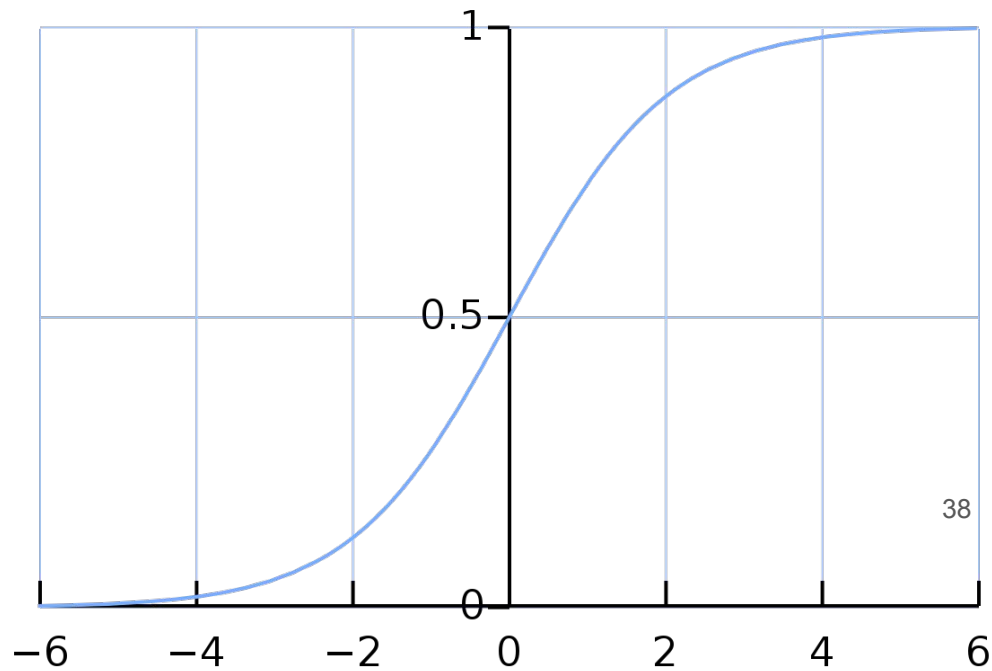
So, a linear model is expressed as a linear function

$$p(X) = w_0 + w_1 X$$

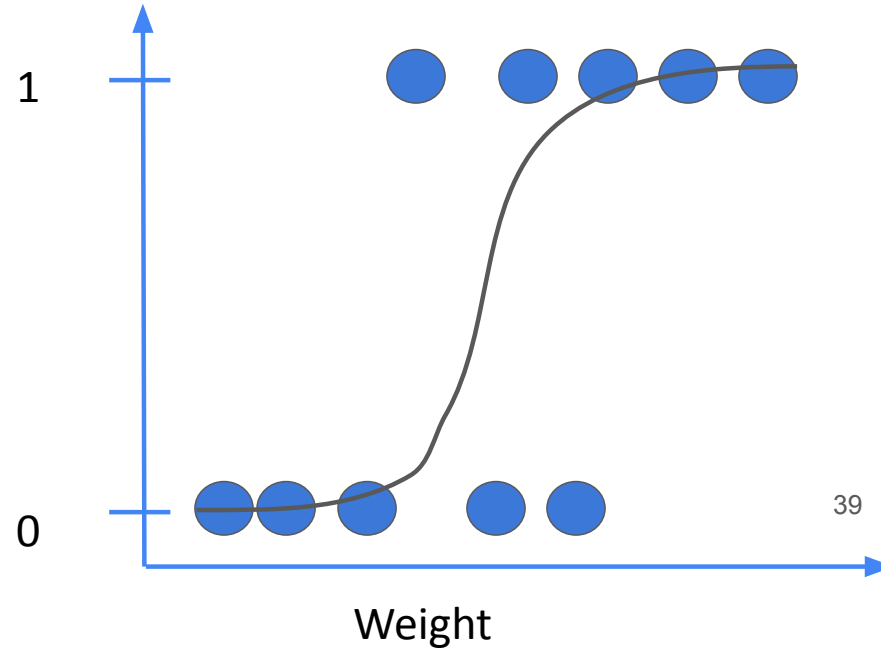
But the logistic model can be expressed as a logistic function a.k.a sigmoid function

$$p(x) = \frac{1}{1 + e^{-x}}$$

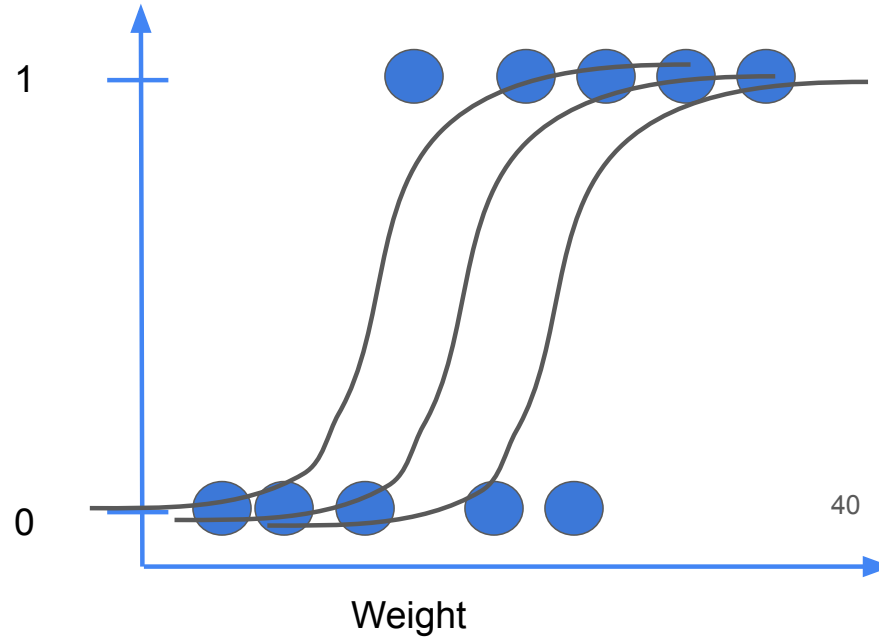
The Sigmoid Graph



We need to fit a sigmoid function like below



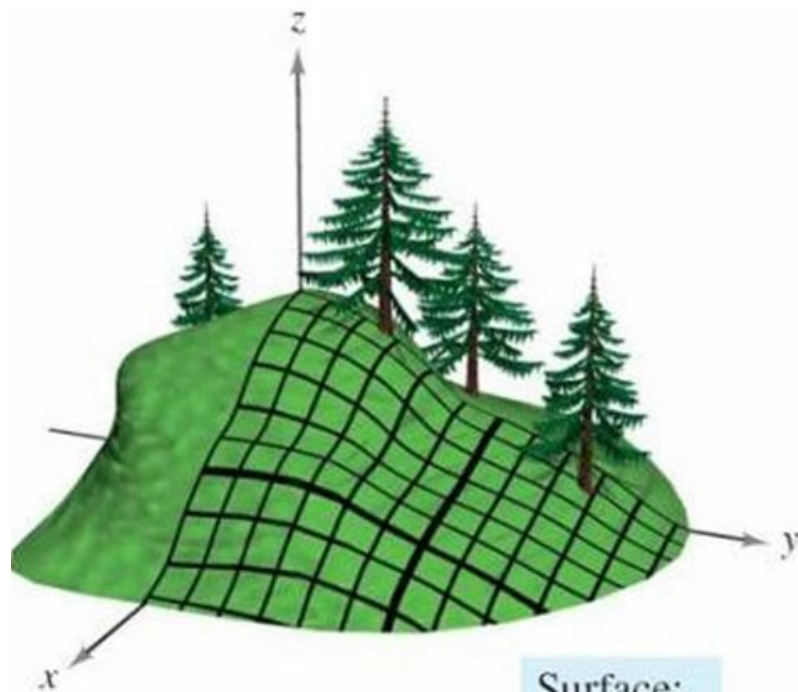
We need to fit a sigmoid function like below



How?

Gradient Descent

Gradient: Geometric Interpretation.



Surface:
 $z = f(x, y)$

He want's to scale the hill:

Let's assume he can take two routes:

One through $x \rightarrow$ co-ordinate(direction)

One through $y \rightarrow$ co-ordinate(direction)

Which route would be fastest?



Whichever direction has highest slope(gradient) i.e.

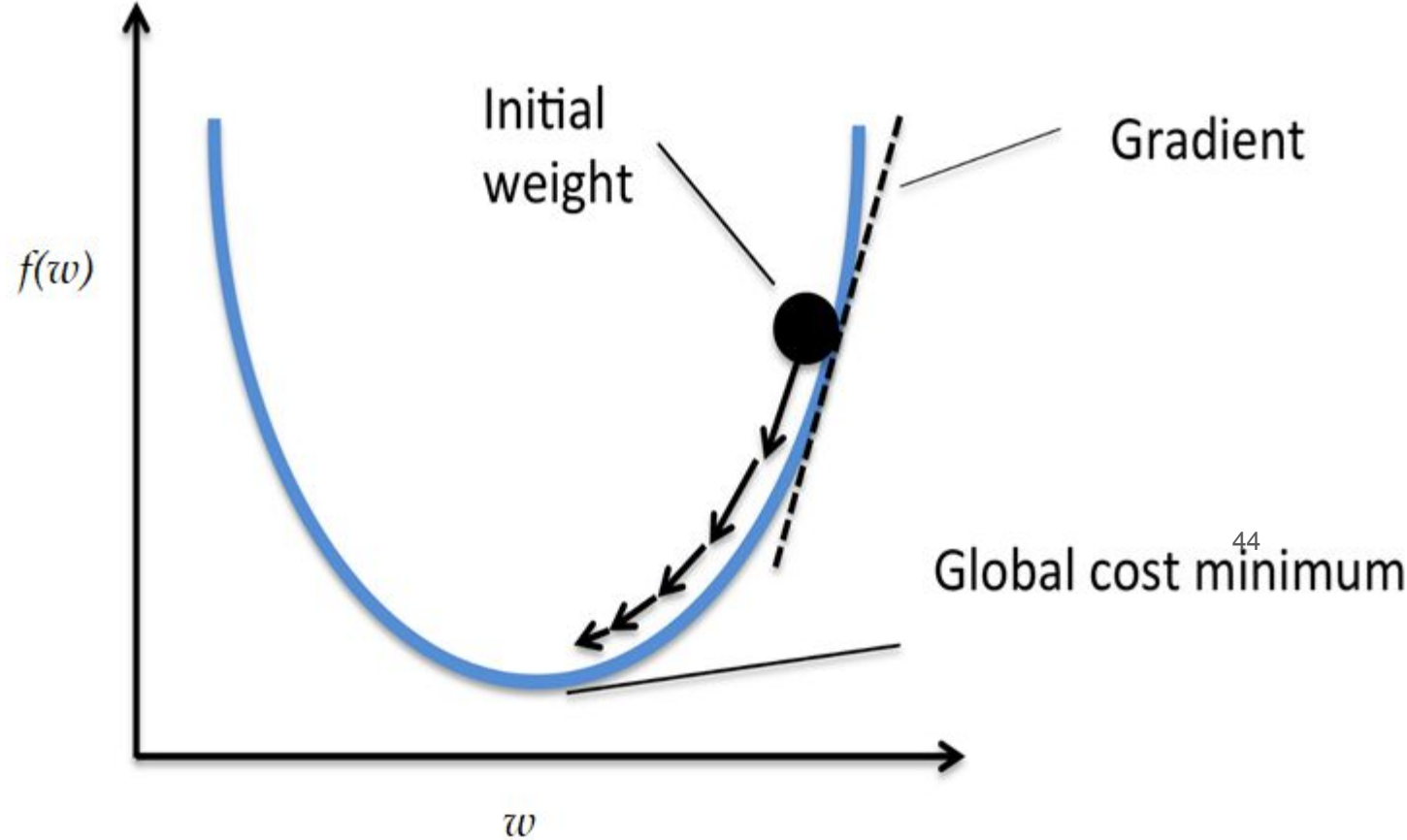
Find the gradient of the surface:

$$z = f(x, y)$$

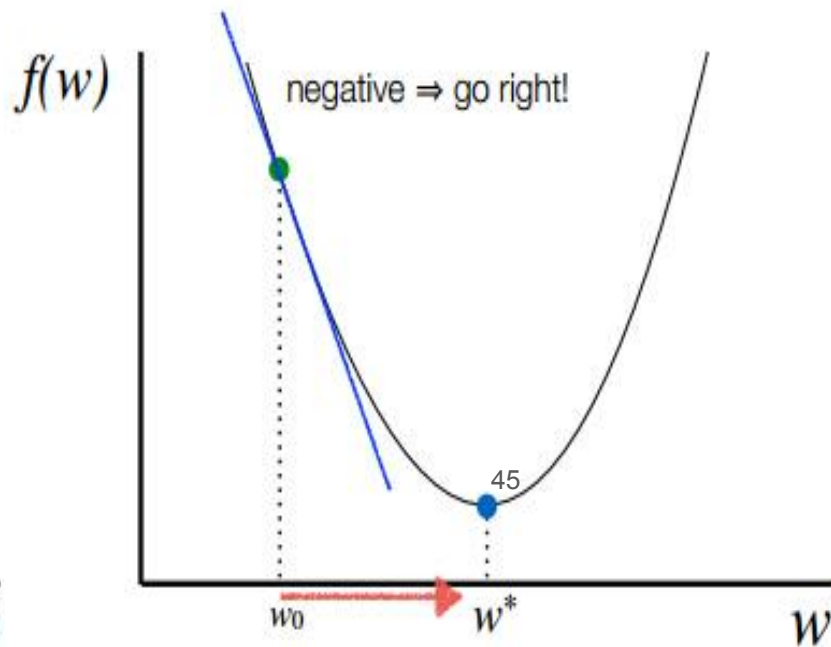
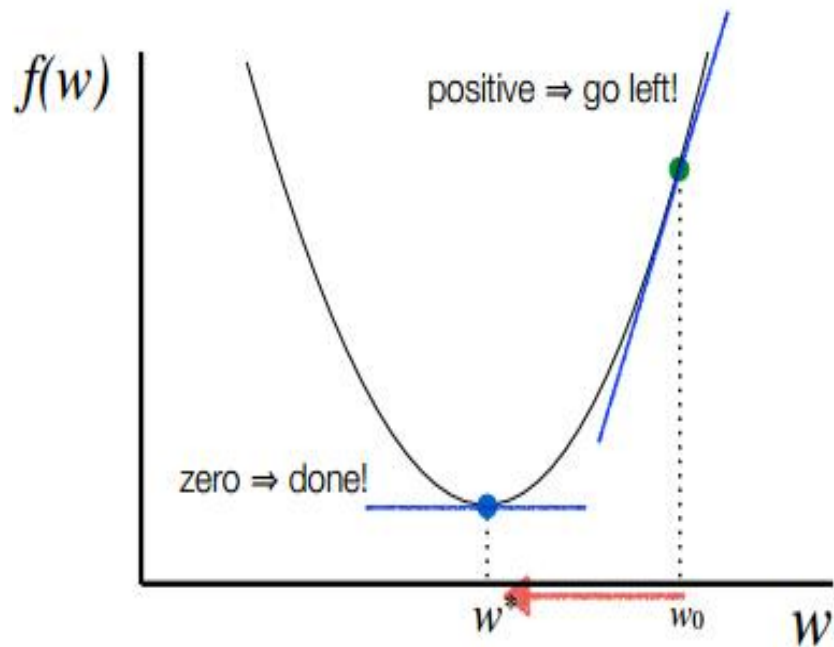
gradient is a partial derivative of z against x and y stack in the vector.

This is read as: "grad. of z " or "grad z " $\leftarrow \nabla z = \left[\frac{\partial f(x, y)}{\partial x} \quad \frac{\partial f(x, y)}{\partial y} \right]$

Gradient Descent



Gradient Descent



Goal?

To minimize $f(w)$ iteratively using the error on the predictions to update the model in such a way as to reduce the error.

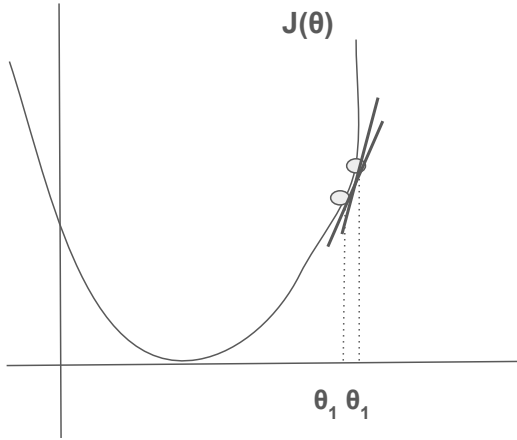
$$w = w - \alpha \frac{df(w, x, y)}{dw}$$

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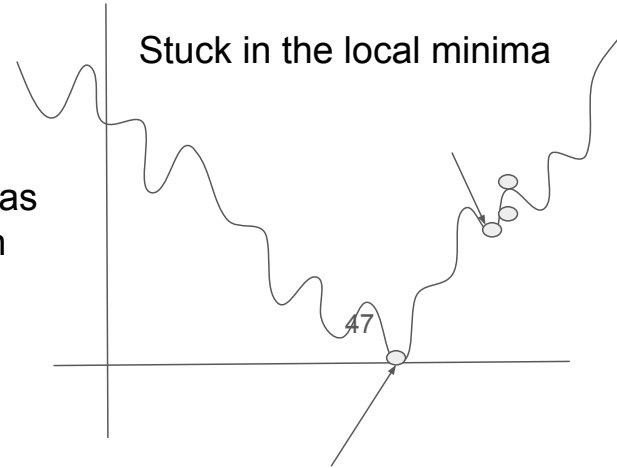
What is $f(w)$?

Cost Function

The cost function for linear regression is $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h(x^{(i)}) - y^{(i)})^2$



But Logistic Regression has
Non-convex loss function



Need to reach here

Cost Function

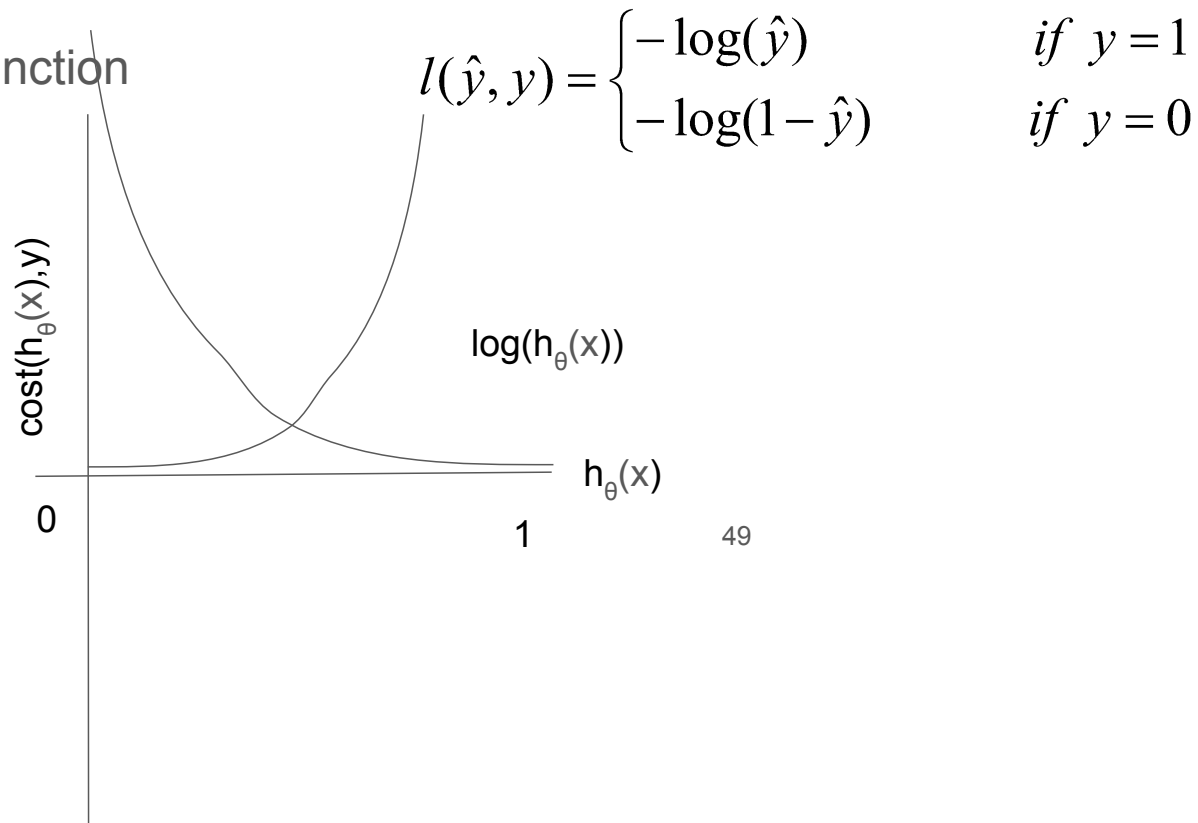
Logistic Regression cost function

$$l(\hat{y}, y) = \begin{cases} -\log(\hat{y}) & \text{if } y = 1 \\ -\log(1 - \hat{y}) & \text{if } y = 0 \end{cases}$$

$$L(y, \hat{y}) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$

Cost Function (Logistic Regression)

Logistic Regression cost function



Cost Function

Logistic Regression cost function

$$l(\hat{y}, y) = \begin{cases} -\log(\hat{y}) & \text{if } y = 1 \\ -\log(1 - \hat{y}) & \text{if } y = 0 \end{cases}$$

$$L(y, \hat{y}) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$

Logistic Regression GD/Derivatives

$$\frac{\partial L(y, \hat{y})}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

$$\frac{\partial L(y, \hat{y})}{\partial y} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial y} = \left\{ -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right\} \{ \hat{y}(1-\hat{y}) \} = \hat{y} - y$$

⊠ if $g(x)$ is logistic function $g'(x) = g(x)(1 - g(x))$

Logistic Regression GD/Derivatives

$$\frac{\partial L}{\partial w_0} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_0} = (\hat{y} - y)$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_1} = x_1(\hat{y} - y)$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_2} = x_2(\hat{y} - y)$$

$$l(\hat{y}, y) = \begin{cases} -\log(\hat{y}) & \text{if } y = 1 \\ -\log(1 - \hat{y}) & \text{if } y = 0 \end{cases}$$

$$\hat{y} = \frac{1}{1 + e^{-Z}}$$

$$Z = w_0 + w_1 x_1 + w_2 x_2$$

Logistic Regression

- Thus, parameters of Logistic Regression are updated as below:

$$w_0 = w_0 - \alpha \frac{\partial L}{\partial w_0}$$

$$w_1 = w_1 - \alpha \frac{\partial L}{\partial w_1}$$

$$w_2 = w_2 - \alpha \frac{\partial L}{\partial w_2}$$

Logistic Regression

- *Example:* Fit the logistic regression model through the following data. Show one epoch of training.

x_1	x_2	$Class(y)$
0.78	0.69	1
0.67	1.00	1
0.00	0.00	0
0.22	0.14	0

Solution

General form of logistic regression equation is: $y = w_0 + w_1x_1 + w_2x_2$

Let us assume that initial values of parameters are:

$$w_0 = w_1 = w_2 = 0$$

Logistic Regression

Iteration 1: $x_1=0.78, x_2=0.69$ $y=1$ $\alpha = 0.1$

$$z = w_0 + w_1x_1 + w_2x_2 = 0$$

$$\hat{y} = \frac{1}{1 + e^{-z}} = 0.5$$

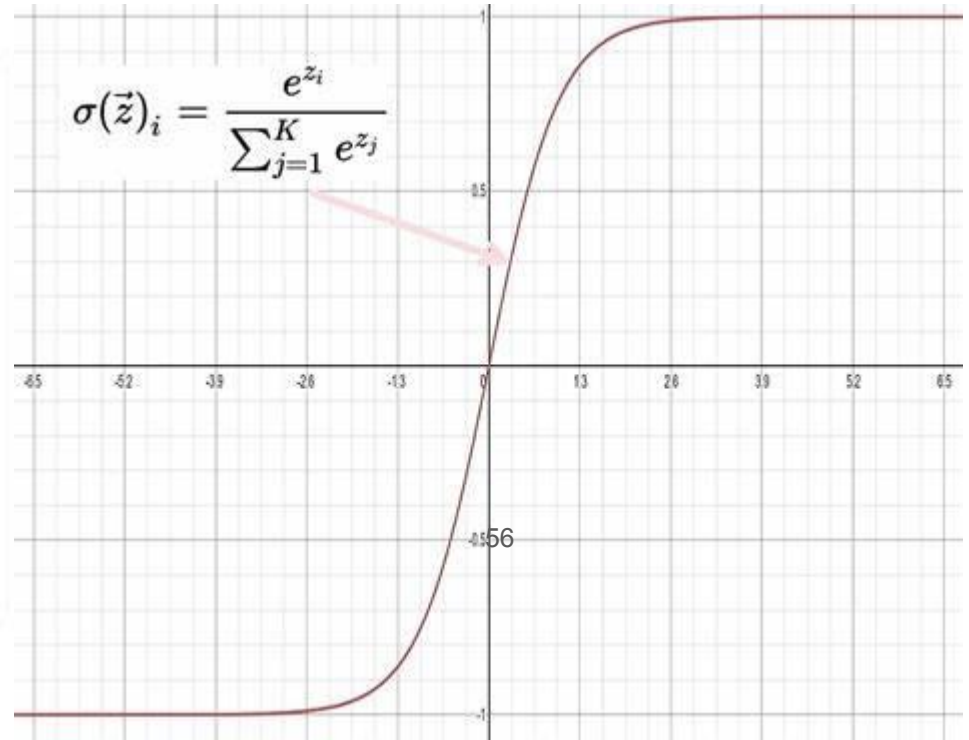
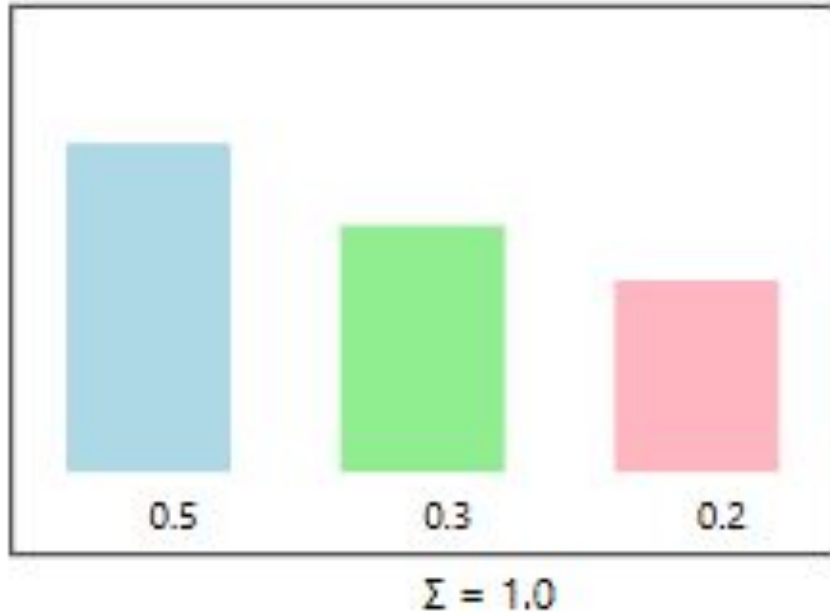
$$w_0 = w_0 - \alpha(\hat{y} - y) = 0 - 0.1 \times (-0.5) = 0.05$$

$$w_1 = w_1 - \alpha(\hat{y} - y)x_1 = 0 - 0.1 \times (-0.5) \times 0.78 = 0.039$$

$$w_2 = w_2 - \alpha(\hat{y} - y)x_2 = 0 - 0.1 \times (-0.5) \times 0.69 = 0.0345$$

In the same way perform iteration 2, 3 and 4.

Multiclass Classification (Soft max Function)



Cost function generalization

$$L(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where:

- y_i is the true label, represented as a one-hot encoded vector⁵⁷.
- $\hat{y}_i$ is the predicted probability for the i -th class.

Derivative of Cost function

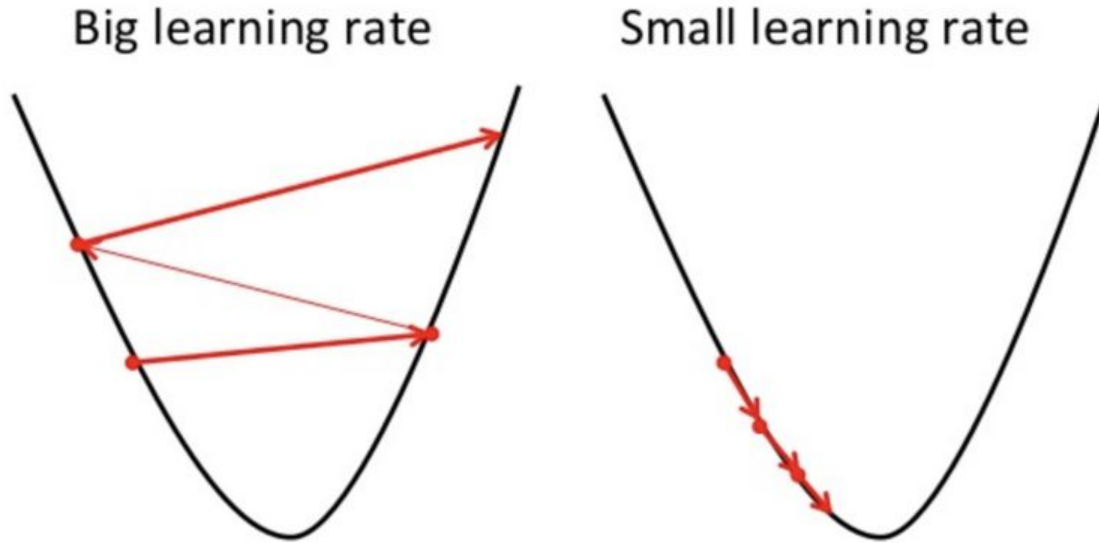
$$\frac{\partial L}{\partial w_{k0}} = \frac{\partial L}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_k} \cdot \frac{\partial z_k}{\partial w_{k0}} = (\hat{y}_k - y_k)$$

$$\frac{\partial L}{\partial w_{k1}} = \frac{\partial L}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_k} \cdot \frac{\partial z_k}{\partial w_{k1}} = x_1(\hat{y}_k - y_k)$$

$$\frac{\partial L}{\partial w_{k2}} = \frac{\partial L}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_k} \cdot \frac{\partial z_k}{\partial w_{k2}} = x_2(\hat{y}_k - y_k)$$

Learning Rate

- How big steps to take?



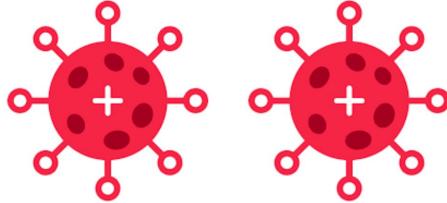
How Good is your Model?

Evaluating Classification Models!

Confusion Matrix

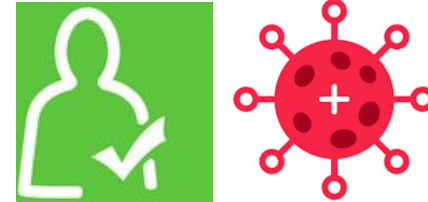
Actually +ve

True Positive



Actually -ve

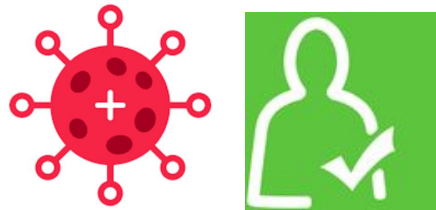
False Positive



Predicted +ve

Predicted -ve

False Negative



True Negative



Confusion Matrix

	Actually +ve	Actually -ve
Predicted +ve	True Positive	False Positive
Predicted -ve	False Negative	True Negative

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$$\text{Accuracy} = (TP+TN) / (TP+TN+FP+FN)$$

Confusion Matrix

	Actually +ve	Actually -ve
Predicted +ve	True Positive	False Positive
Predicted -ve	False Negative	True Negative

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$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Confusion Matrix

	Actually +ve	Actually -ve
Predicted +ve	True Positive	False Positive
Predicted -ve	False Negative	True Negative

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$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Confusion Matrix

	Actually +ve	Actually -ve
Predicted +ve	True Positive	False Positive
Predicted -ve	False Negative	True Negative

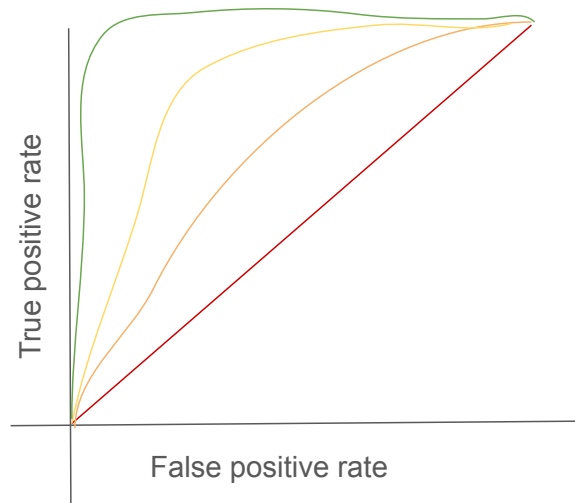
65

$$\text{F1 Score} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$

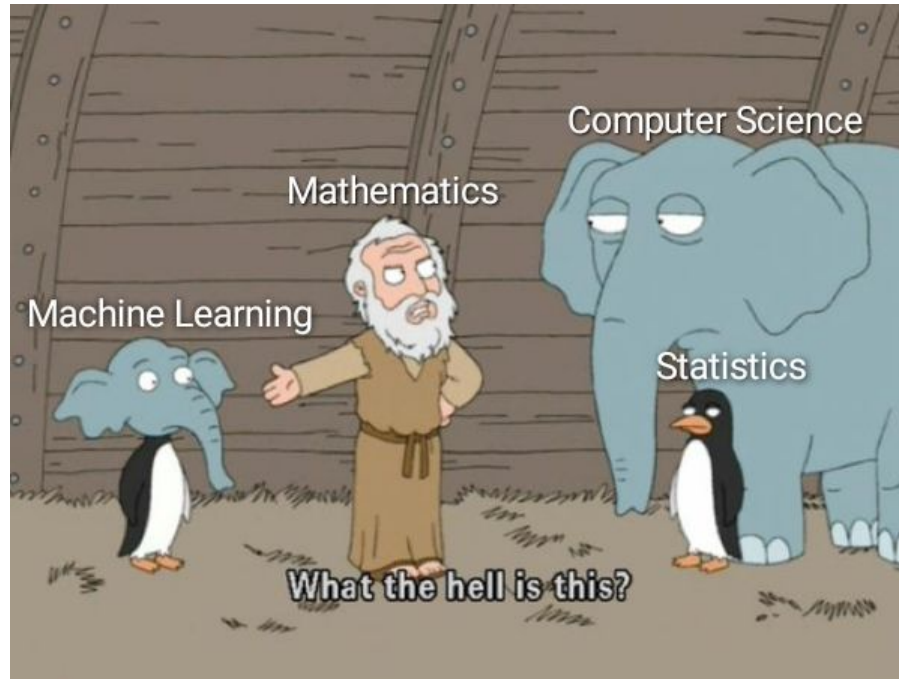
AUC-ROC Curve

How good the model can predict each classes

ROC (Receiver Operating Characteristic) curve is a visualization of false positive rate (x-axis) and the true positive rate (y-axis).



Discussion?



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Thank You!